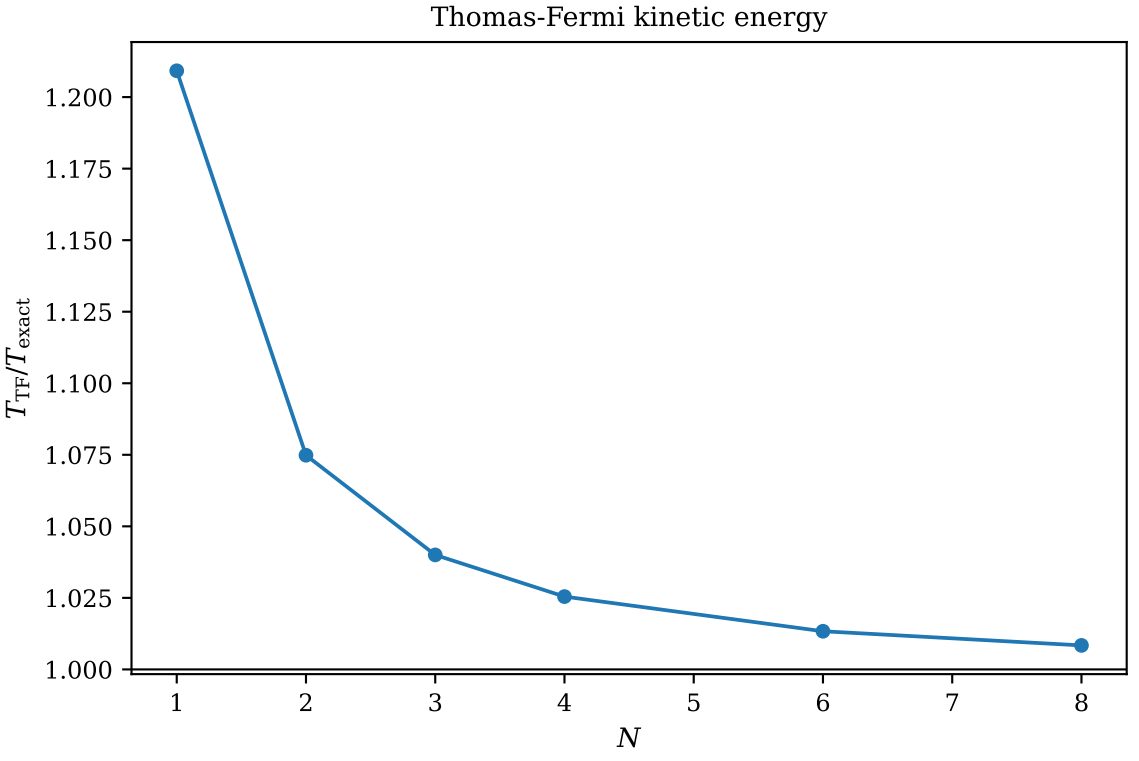
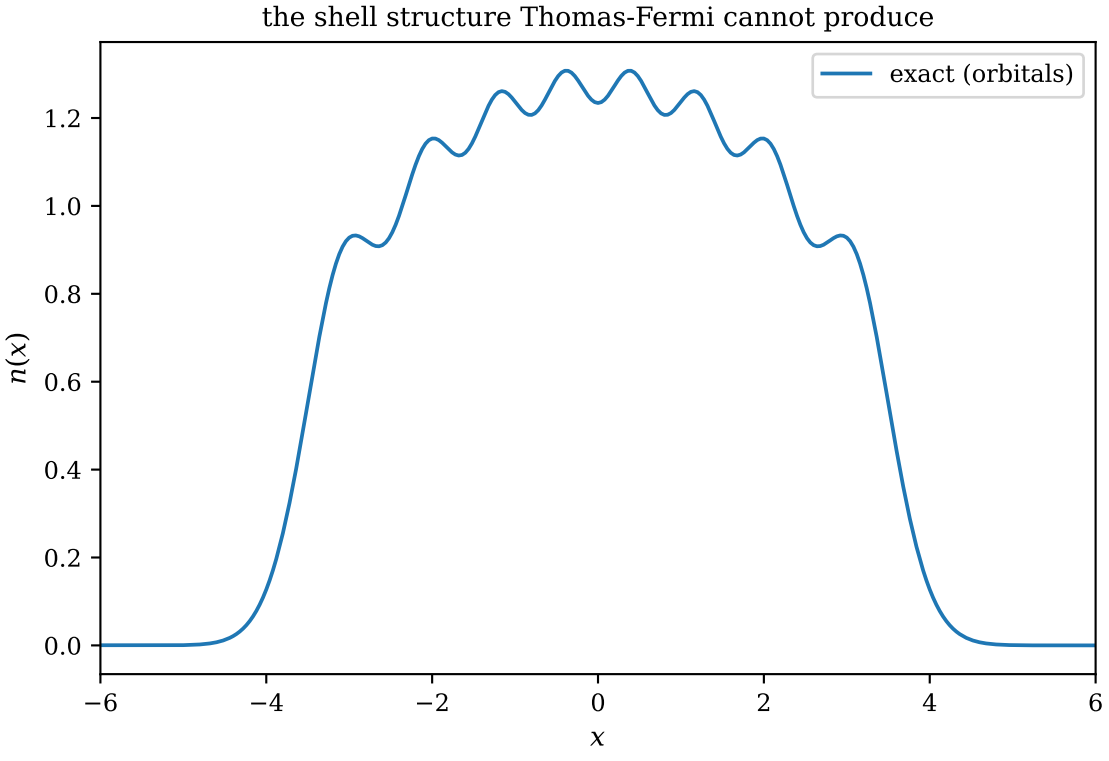

Thomas-Fermi kinetic energy
$T_{\mathrm{TF}}/T_{\mathrm{exact}}$
1.200
1.175
1.150
1.125
1.100
1.075
1.050
1.025
1.000
1
2
3
4
5
6
7
8
N


the shell structure Thomas-Fermi cannot produce
exact (orbitals)
$n(x)$
1.2
1.0
0.8
0.6
0.4
0.2
0.0
−6
−4
−2
0
2
4
6
x